Yuan Yuan

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Education

University of California, Los Angeles Sep 2026 – May 2031 (Expected)

- Ph.D. in Computer Science
  Advisor: Nanyun Peng

University of Pennsylvania Sep 2023 – May 2026

- M.S. Data Science & Computer and Information Science GPA: 3.88/4.0

Vassar College Sep 2017 – May 2021

- B.A., Major in Political Science, Correlate in Math GPA: 3.93/4.0

Professional Experience

AI Engineer, Microsoft – Remote Sep 2025 – Dec 2025

- Developed and debugged **inference pipelines** for open- and closed-source LLMs, monitoring performance on personalization benchmarks.
- Maintained an internal multi-party **LLM chat platform** built with **FastAPI** and **HTML** for real-time collaboration.
- **Deployed and integrated new models** into the platform, ensuring seamless user interaction and stability.
- Extended model capabilities by implementing **GPT-style function calling** for enhanced reasoning and tool use.

Software Engineer Intern, Meta – Seattle, US May 2025 – Aug 2025

- Built and shipped features for the **Brand Safety Advertiser Experience Team**, collaborating with cross-functional product designers, PMs, and engineers.
- **Initiated and scoped the first AI Agent project** for the team; designed and implemented an internal **Oncall Domain Agent** using Hack and Context Engineering, improving oncall efficiency and developer productivity.
- Enhanced **millions** of advertisers' experience with Facebook Ads by **improving UIs in JavaScript** and **streamlining workflows**, significantly reducing friction in ad interactions.
- Developed an **ads matcher in Hack/PHP** to automatically detect and filter ads based on advertiser requirements, increasing accuracy of ad delivery and covering over **20 million** ads on platforms.

Selected Research

Multi-Agent Inference: Coordination, Exploration, and Reliable Evaluation
Work in Progress

- Built a **multi-agent inference framework** integrating native Codex and Claude Code harnesses with local Qwen models served through vLLM and hosted DeepSeek models. Implemented four-agent shared-memory, independent, and coordinator-led workflows with dynamic delegation, persistent sessions, and controlled candidate access.
- Designed a **12-condition KernelBench LeNet-5 study** across two models, two harnesses, and three coordination architectures, with fixed one-hour runs, token accounting, and reproducible evaluation. Audited **3,764 tool calls across six orchestrator runs** and replayed all winning submissions, uncovering evaluator-source leakage and incomplete correctness checks that confounded apparent performance gains.

***Yuan Yuan**, Haoyi Qiu, Jason Wu, Pranav Narayanan Venkit, Nanyun Peng.*

PersonaMem-v2: Towards Personalized Intelligence via Learning Implicit User Personas and Agentic Memory
KDD 2027

- We built a **synthetic dataset** to evaluate and improve LLMs' ability to detect **implicit user preferences** in long-term conversations; using **Reinforcement Fine-Tuning (RFT)**, we design reward signals that encourage models to infer subtle, unspoken preferences and adapt responses accordingly. Our results show that RFT-trained models achieve stronger personalization and alignment with evolving user needs, advancing beyond memory evaluation toward more adaptive, context-aware AI assistants.
- Trained Qwen3-4B with reinforcement fine-tuning using VERL for implicit preference inference; developed an agentic memory system that maintains a compact, human-readable user memory across long-term conversations.

*Bowen Jiang, **Yuan Yuan**, Maohao Shen, Zhuoqun Hao, Zhangchen Xu, Zichen Chen, Ziyi Liu, Anvesh Rao Vijjini, Jiashu He, Hanchao Yu, Radha Poovendran, Gregory Wornell, Lyle Ungar, Dan Roth, Sihao Chen, Camillo Jose Taylor.*

Know Me, Respond to Me: Benchmarking LLMs for Dynamic User Profiling and Personalized Responses at Scale
COLM 2025 & Long abstract at the MASC-SLL 2025 [Oral]
- We propose **PersonaMem**, a benchmark built from over 180 simulated user–LLM interaction histories (each with up to 60 sessions over 15 tasks) that captures evolving user personas over time. Given an in-situ user query, models must choose responses aligned with the user's current profile state. We find that even leading LLMs struggle: they often fail to adapt to dynamic profile changes, achieving only ∼50% accuracy overall.
- Evaluated GPT, Qwen, Llama, and Claude models on the benchmark.

Bowen Jiang, Zhuoqun Hao*, Young-Min Cho, Bryan Li, **Yuan Yuan**, Sihao Chen, Lyle Ungar, Camillo J. Taylor*, Dan Roth*.*

Publications

Real2Sim Bench: Evaluating Metric 3D Reconstruction from Real-World Images and Video
Work in Progress
*Yu Zhou, **Yuan Yuan**, Cheng-fu Yang, Di Wu, Rui Sun, Zhe Gan, Kaiwei Chang, Nanyun Peng.*

A Concise Agent is Less Expert: Revealing Side Effects of Using Style Features on Conversational Agents
Under Review of ARR January
Young-Min Cho, **Yuan Yuan***, Sharath Chandra Guntuku, Lyle Ungar.*

AgentWorld: Benchmarking Long-Horizon Collaboration of Multi-agent LLMs
COLM 2026
*Raphael Shu, Yusen Zhang, Young Min Cho, Jin Mo Yang, **Yuan Yuan**, Wenliang Zheng, Sharath Chandra Guntuku, Lyle Ungar, Zhou Yu, Rui Zhang.*

One Model, All Roles: Multi-Turn, Multi-Agent Self-Play Reinforcement Learning for Conversational Social Intelligence
arXiv preprint arXiv:2602.03109, 2026
*Bowen Jiang, Taiwei Shi, Ryo Kamoi, **Yuan Yuan**, Camillo J. Taylor, Longqi Yang, Pei Zhou, Sihao Chen.*

TurnaboutLLM: A Deductive Reasoning Benchmark from Detective Games
EMNLP 2025 Main
Yuan Yuan, Muyu He*, Muhammad Adil Shahid, Jiani Huang, Ziyang Li, Li Zhang.*

ControlText: Unlocking Controllable Fonts in Multilingual Text Rendering without Font Annotations
EMNLP 2025 Findings
Bowen Jiang, **Yuan Yuan***, Xinyi Bai, Zhuoqun Hao, Alyson Yin, Yaojie Hu, Wenyu Liao, Lyle Ungar, Camillo J. Taylor.*

Multilingual Retrieval Augmented Generation for Culturally-Sensitive Tasks: A Benchmark for Cross-lingual Robustness
ACL 2025 Findings
Bryan Li, Fiona Luo, Samar Haider*, Adwait Agashe*, Tammy Li†, Runqi Liu†, Muqing Miao†, Shriya Ramakrishnan†, **Yuan Yuan**†, Chris Callison-Burch.*

Towards Rationality in Language and Multimodal Agents: A Survey
NAACL 2025 Main
*Bowen Jiang, Yangxinyu Xie, Xiaomeng Wang, **Yuan Yuan**, Zhuoqun Hao, Xinyi Bai, Weijie J. Su, Camillo J. Taylor, Tanwi Mallick.*

Open-Source Contributions

AsyncAPI (Google Summer of Code Maintainer): Led the migration of the Generator CLI and integrated modular templates to enhance developer experience.
- **PRs:** #1251, #1234, #1213 **Issue:** #1249

vLLM (Contributor): Participated in architectural discussions regarding distributed inference reliability and multi-node synchronization.

- **Issue:** #33034 (Proposed a benchmark-driven auto-tuning solution to replace hard-coded Mamba kernel configurations with a dynamic JSON-based loader)

Honors

PHI BETA KAPPA Membership
The Julia Flitner Lamb Prize at Vassar College
Ann Cornelisen Fund for Post-Graduate Fellowships

TECHNICAL SKILLS & FRAMEWORKS

Java, Python, R, C++, Spring Boot, Django, AWS EC2, AWS S3, AWS Kinesis, AWS RDS, AWS Aurora, AWS Elasticache, Redis, Apache Kafka, MySQL, Postgres, DynamoDB, MongoDB, Docker, Docker-compose, Kubernetes, Jenkins, ArgoCD, gRPC, HTTP, WebSocket